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Acronyms

I try hard to avoid using acronyms to reduce the unnecessary burden for reading. However, the following acronyms are standard and will be used repeatedly.

ANOVA	(Fisher's) analysis of variance
BLUE	best linear unbiased estimator (in Gauss–Markov Theorem)
CATE	conditional average treatment effect
CDF	cumulative distribution function
CLT	central limit theorem
CV	cross-validation
EHW	Eicker–Huber–White (robust covariance matrix or standard error)
FDA	U.S. Food and Drug Administration
FWL	Frisch–Waugh–Lovell (theorem)
GEE	generalized estimating equation
GLM	generalized linear model
HC	heteroskedasticity-consistent (covariance matrix or standard error)
IID	independent and identically distributed
LAD	least absolute deviations
lasso	least absolute shrinkage and selection operator
MLE	maximum likelihood estimate
OLS	ordinary least squares
PCA	principal component analysis
RCT	randomized controlled trial
RSS	residual sum of squares
SVD	singular value decomposition
WLS	weighted least squares

Symbols

All vectors are column vectors as in R unless stated otherwise. Let the superscript "T" denote the transpose of a vector or matrix.

$\mathbb{R}$	the set of all real numbers
$\mathbb{R}^p$	the set of p-dimensional vectors with real components
$\perp\!\!\!\perp$	independence and conditional independence
$\overset{\text{IID}}{\sim}$	independent and identically distributed (IID)
$\overset{\text{a}}{\sim}$	approximation in distribution
I_n	identity matrix of dimension $n \times n$
x_i	covariate vector for unit i
y_i	outcome for unit i
X	covariate matrix
Y	outcome vector
H	hat matrix $H = X(X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}$
h_{ii}	leverage score: the $(i, i)$the element of the hat matrix H
β	regression coefficient
ε	error term

Useful R packages

This book uses the following R packages and functions.

package	function or data	use
car	hccm	Eicker–Huber–White robust standard error
	linearHypothesis	testing linear hypotheses in linear models
foreign	read.dta	read stata data
gee	gee	Generalized estimating equation
HistData	GaltonFamilies	Galton's data on parents' and children's heights
MASS	lm.ridge	ridge regression
	glm.nb	Negative-Binomial regression
glmnet	cv.glmnet	Lasso with cross-validation
mlbench	BostonHousing	Boston housing data
	polr	proportional odds logistic regression
Matching	lalonde	LaLonde data
nnet	multinom	Multinomial logistic regression
quantreg	rq	quantile regression
survival	coxph	Cox proportional hazards regression
	survdiff	log rank test
	survfit	Kaplan–Meier curve
ElemStatLearn	prostate	data for Hastie et al. (2009)
wooldridge	mroz	data for Wooldridge (2016)

1

Motivations for Statistical Models

This book is about the linear model and its extensions. Before delving into the mathematical details of specific models, I will briefly provide some motivations for studying statistical models.

1.1 Data and statistical models

A wide range of problems in statistics and machine learning have the following data structure:

Unit	outcome/response	covariates/features/predictors			
i	Y	X_1	X_2	$\cdots$	X_p
1	y_1	x_{11}	x_{12}	$\cdots$	x_{1p}
2	y_2	x_{21}	x_{22}	$\cdots$	x_{2p}
$\vdots$	$\vdots$	$\vdots$	$\vdots$		$\vdots$
n	y_n	x_{n1}	x_{n2}	$\cdots$	x_{np}

For each unit i, we observe the outcome of interest (also called the response), y_i, as well as p covariates (also called features or predictors), $x_{i1}, \ldots, x_{ip}$. We often use

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

to denote the n-dimensional outcome vector, and

$$X = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}$$

to denote the $n \times p$ covariate matrix, also called the *design matrix*. In most cases, the first column of X contains constants 1s.

Based on the data (X, Y), we can ask the following questions:

(Q1) Describe the relationship between X and Y, i.e., their association or correlation. For example, how is the patients' average height related to the children's average height? How is one's height related to one's weight? How are one's education and working experience related to one's income?

(Q2) Predict Y^* with new data X^*, based on the old data (X, Y). In particular, we want to use the current data (X, Y) to train a predictor, and then use it to predict future Y^* based on future X^*. This is called *supervised learning* in the field of machine learning. For example, how do we predict whether an email is spam or not based on the frequencies of the most commonly occurring words and punctuation marks in the email? How do we predict cancer patients' survival time based on their clinical measures?

(Q3) Estimate the causal effect of some components in X on Y. What if we change some components of X? How do we measure the impact of the hypothetical intervention of some components of X on Y? This is a much harder question because most statistical tools are designed to infer association, not causation. For example, the U.S. Food and Drug Administration (FDA) approves drugs based on randomized controlled trials (RCT) because RCTs are most credible to infer causal effects of drugs on health outcomes. Economists are interested in evaluating the effect of a job training program on employment and wages. However, this is a notoriously difficult problem because participation in the job training program is not randomized in observational data.

The above descriptions are about generic X and Y, which can have many different types. We often use different statistical models to capture the features of different types of data. Below I give a brief overview of models that will appear in later parts of this book.

(T1) X and Y are univariate and continuous. In Francis Galton's[1] classic example, X is the parents' average height and Y is the children's average height (Galton, 1886). Let $\hat{y}_i$ denote the "fitted value" of the outcome for unit i with covariate value x_i. Galton derived the following formula:

$$\hat{y}_i = \bar{y} + \hat{\rho}\frac{\hat{\sigma}_y}{\hat{\sigma}_x}(x_i - \bar{x})$$

which is equivalent to

$$\frac{\hat{y} - \bar{y}}{\hat{\sigma}_y} = \hat{\rho}\frac{x_i - \bar{x}}{\hat{\sigma}_x}, \tag{1.1}$$

where

$$\bar{x} = n^{-1}\sum_{i=1}^{n} x_i, \qquad \bar{y} = n^{-1}\sum_{i=1}^{n} y_i$$

are the sample means,

$$\hat{\sigma}_x^2 = (n-1)^{-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad \hat{\sigma}_y^2 = (n-1)^{-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$$

are the sample variances, and $\hat{\rho} = \hat{\sigma}_{xy}/(\hat{\sigma}_x\hat{\sigma}_y)$ is the sample Pearson correlation coefficient with the sample covariance

$$\hat{\sigma}_{xy} = (n-1)^{-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}).$$

The identity (1.1) is the famous formula of "regression towards mediocrity" or "regression towards the mean". Galton first introduced the terminology "regression."[2] Galton

[1] Who was Francis Galton? He was Charles Darwin's nephew and was famous for his pioneer work in statistics and for devising a method for classifying fingerprints that proved useful in forensic science. He also invented the term *eugenics*, a field that causes a lot of controversies nowadays.

[2] The name "regression" is widely used in statistics now. For instance, we sometimes use "linear regression" interchangeably with "linear model." We also extend the name to "logistic regression" or "Cox regression," which will be discussed in later chapters of this book.

called regression because the relative deviation of the children's average height is smaller than that of the parents' average height if $|\hat{\rho}| < 1$. We will derive (1.1) in Chapter 2.

(T2) Y is univariate and continuous, and X is multivariate of mixed types. In the `R` package `ElemStatLearn`, the dataset `prostate` has an outcome of interest as the log of the prostate-specific antigen `lpsa` and some potential predictors including the log cancer volume `lcavol`, the log prostate weight `lweight`, age `age`, etc. The main chapters of this book, Chapters 3–19, will discuss linear regression with multidimensional covariates.

(T3) Y is binary or indicator of two classes, and X is multivariate of mixed types. For example, in the `R` package `wooldridge`, the dataset `mroz` contains an outcome of interest being the binary indicator for whether a woman was in the labor force in the year 1975, and some useful covariates are in the table below:

covariate name	covariate meaning
kidslt6	number of kids younger than six years old
kidsge6	number of kids between six and eighteen years old
age	age
educ	years of education
husage	husband's age
huseduc	husband's years of education

Chapter 20 will discuss *logistic regression* for binary outcomes.

(T4) Y is categorical without ordering. For example, the choice of housing type, single-family house, townhouse, or condominium, is a categorical variable. Chapter 21 will discuss *multinomial logistic regression* for categorical outcomes without ordering.

(T5) Y is categorical and ordered. For example, the final course evaluation at UC Berkeley can take value in $\{1, 2, 3, 4, 5, 6, 7\}$. These numbers have clear ordering but they are not the usual real numbers. Chapter 21 will discuss *proportional odds regression* for ordered categorical outcomes.

(T6) Y represents counts. For example, the number of times one went to the gym last week is a non-negative integer representing counts. Chapter 22 will discuss regression models for count outcomes.

(T7) Y is multivariate and correlated. In medical trials, the data are often longitudinal, meaning that the patient's outcomes are measured repeatedly over time. So each patient has a multivariate outcome. In field experiments of public health and development economics, the randomized interventions are often at the village level but the outcome data are collected at the household level. So within villages, the outcomes are correlated. Chapter 25 will discuss the *generalized estimating equation* for correlated data.

(T8) Y represent time-to-event outcomes. For example, in medical trials, a major outcome of interest is the survival time; in labor economics, a major outcome of interest is the time to find the next job. The former is called *survival analysis* in biostatistics and the latter is called *duration analysis* in econometrics. Chapter 27 will discuss *Cox proportional hazards regression.*

1.2 Why linear models?

Why do we study linear models if many real problems may have nonlinear structures? There are important reasons.

(R1) Linear models are simple but non-trivial starting points for learning.

(R2) Linear models can provide insights because we can derive explicit formulas based on elegant algebra and geometry.

(R3) Linear models can handle nonlinearity by incorporating nonlinear terms of covariates, for example, X can contain the polynomials or nonlinear transformations of the original covariates. In statistics, "linear" means *linear in parameters*, not *linear in covariates*.

(R4) Linear models can be good approximations of nonlinear data-generating processes.

(R5) Linear models are simpler than nonlinear models, but they do not necessarily perform worse than more complicated nonlinear models. We have finite data so we cannot fit arbitrarily complicated models.

If you are interested in nonlinear models, you can read another book on machine learning.

20

Logistic Regression for Binary Outcomes

Many applications have binary outcomes $y_i \in \{0, 1\}$. This chapter discusses statistical models of binary outcomes, focusing on the logistic regression, also called the logit regression for simplicity.

20.1 Regression with binary outcomes

20.1.1 Linear probability model

For simplicity, we can still use the linear model for a binary outcome. It is also called the *linear probability model*:

$$y_i = x_i^{\mathrm{T}}\beta + \varepsilon_i \qquad \text{with} \qquad E(\varepsilon_i \mid x_i) = 0$$

because the conditional probability of $y_i = 1$ given x_i is a linear function of x_i:

$$\mathrm{pr}(y_i = 1 \mid x_i) = E(y_i \mid x_i) = x_i^{\mathrm{T}}\beta.$$

An advantage of this linear model is that the interpretation of the coefficient remains the same as linear models for general outcomes:

$$\frac{\partial \mathrm{pr}(y_i = 1 \mid x_i)}{\partial x_{ij}} = \beta_j,$$

that is, β_j measures the partial impact of x_{ij} on the probability of y_i.

A minor technical issue is that the linear probability model implies heteroskedasticity because

$$\begin{aligned} \mathrm{var}(y_i \mid x_i) &= \mathrm{pr}(y_i = 1 \mid x_i)(1 - \mathrm{pr}(y_i = 1 \mid x_i)) \\ &= x_i^{\mathrm{T}}\beta(1 - x_i^{\mathrm{T}}\beta). \end{aligned}$$

Therefore, we must use the EHW covariance based on OLS. We can also use the feasible generalized least squares (FGLS) in Chapter 19.3.1 to improve efficiency over OLS.

A more severe problem with the linear probability model is its plausibility in general. We may not believe that a linear model is the correct model for a binary outcome because the probability $\mathrm{pr}(y_i = 1 \mid x_i)$ on the left-hand side is bounded between 0 and 1, but the linear combination $x_i^{\mathrm{T}}\beta$ on the right-hand side can be unbounded for general covariates and coefficients. Nevertheless, the OLS decomposition $y_i = x_i^{\mathrm{T}}\beta + \varepsilon_i$ works for any $y_i \in \mathbb{R}$, so it is applicable for binary y_i.

Sometimes, practitioners feel that the linear model is not natural for binary outcomes because the predicted value can be outside the range of $[0, 1]$. Therefore, it is more reasonable to build a model that automatically accommodates the binary feature of the outcome.

20.1.2 General link functions

A linear combination of general covariates may be outside the range of $[0, 1]$, but we can find a monotone transformation to force it to lie within the interval $[0, 1]$. This motivates us to consider the following model:

$$\text{pr}(y_i = 1 \mid x_i) = g(x_i^{\text{T}}\beta),$$

where $g(\cdot) : \mathbb{R} \to [0, 1]$ is a monotone function, and its inverse is often called the link function. Mathematically, the distribution function of any continuous random variable is a monotone function that maps from $\mathbb{R}$ to $[0, 1]$. So we have infinitely many choices for $g(\cdot)$. Four canonical choices "logit", "probit", "cauchit", and "cloglog" are below which are the standard options in R:

name	functional form
logit	$g(z) = \frac{e^z}{1+e^z}$
probit	$g(z) = \Phi(z)$
cauchit	$g(z) = \frac{1}{\pi}\arctan(z) + \frac{1}{2}$
cloglog	$g(z) = 1 - \exp(-e^z)$

The above $g(z)$'s correspond to different distribution functions. The $g(z)$ for the logit model[1] is the distribution function of the standard logistic distribution with density

$$g'(z) = \frac{e^z}{(1 + e^z)^2} = g(z)\left\{1 - g(z)\right\}. \tag{20.1}$$

The $g(z)$ for the probit model[2] is the distribution function of a standard Normal distribution. The $g(z)$ for the cauchit model is the distribution function of the standard Cauchy distribution with density

$$g'(z) = \frac{1}{\pi(1 + z^2)}.$$

The $g(z)$ for the cloglog model is the distribution function of the standard log-Weilbull distribution with density

$$g'(z) = \exp(z - e^z).$$

I will give more motivations for the first three link functions in Section 20.7.1 and for the fourth link function in Problem 22.6.

Figure 20.1 shows the distributions and densities of the corresponding link functions. The distribution functions are quite similar for all links, but the density for cloglog is asymmetric although all other three densities are symmetric.

This chapter will focus on the logit model, and extensions to other models are conceptually straightforward. We can also write the logit model as

$$\text{pr}(y_i = 1 \mid x_i) \equiv \pi(x_i, \beta) = \frac{e^{x_i^{\text{T}}\beta}}{1 + e^{x_i^{\text{T}}\beta}}, \tag{20.2}$$

for the conditional probability of y_i given x_i, or, equivalently,

$$\text{logit}\left\{\text{pr}(y_i = 1 \mid x_i)\right\} \equiv \log \frac{\text{pr}(y_i = 1 \mid x_i)}{1 - \text{pr}(y_i = 1 \mid x_i)} = x_i^{\text{T}}\beta,$$

for the log of the odds of y_i given x_i, with the logit function

$$\text{logit}(\pi) = \log \frac{\pi}{1 - \pi}.$$

[1] Berkson (1944) was an early use of the logit model.
[2] Bliss (1934) was an early use of the probit model.

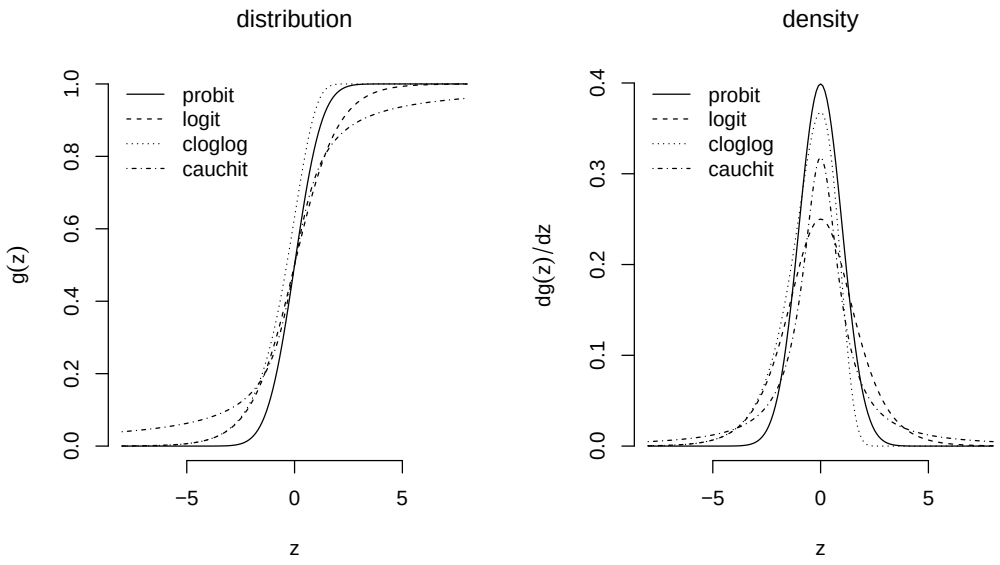

FIGURE 20.1: Distributions and densities corresponding to the link functions

Because y_i is a binary random variable, its probability completely determines its distribution. So we can also write the logit model in the following form:

Assumption 20.1 (binary logistic regression model) *We have*

$$y_i \mid x_i \sim \textit{Bernoulli}\left(\frac{e^{x_i^{\mathrm{T}}\beta}}{1+e^{x_i^{\mathrm{T}}\beta}}\right).$$

The observations are independent across units. The β is the unknown parameter.

Each coefficient β_j measures the impact of x_{ij} on the log odds of the outcome:

$$\frac{\partial}{\partial x_{ij}}\text{logit}\{\text{pr}(y_i = 1 \mid x_i)\} = \beta_j.$$

Epidemiologists also call β_j the conditional log odds ratio[3] because

$$\begin{aligned}
\beta_j &= \text{logit}\left\{\text{pr}(y_i = 1 \mid \ldots, x_{ij}+1, \ldots)\right\} - \text{logit}\left\{\text{pr}(y_i = 1 \mid \ldots, x_{ij}, \ldots)\right\} \\
&= \log \frac{\text{pr}(y_i = 1 \mid \ldots, x_{ij}+1)}{1-\text{pr}(y_i = 1 \mid \ldots, x_{ij}+1)} - \log \frac{\text{pr}(y_i = 1 \mid \ldots, x_{ij}, \ldots)}{1-\text{pr}(y_i = 1 \mid \ldots, x_{ij}, \ldots)} \\
&= \log \left\{ \frac{\text{pr}(y_i = 1 \mid \ldots, x_{ij}+1, \ldots)}{1-\text{pr}(y_i = 1 \mid \ldots, x_{ij}+1, \ldots)} \Big/ \frac{\text{pr}(y_i = 1 \mid \ldots, x_{ij}, \ldots)}{1-\text{pr}(y_i = 1 \mid \ldots, x_{ij}, \ldots)} \right\},
\end{aligned}$$

that is, the change of the log odds of y_i if we increase x_{ij} by a unit holding other covariates unchanged. Qualitatively, if $\beta_j > 0$, then larger values of x_{ij} lead to larger probabilities of $y_i = 1$; if $\beta_j < 0$, then larger values of x_{ij} lead to smaller probabilities of $y_i = 1$.

[3]In probability theory, if p is the probability, then $p/(1-p)$ is called the odds. It is a terminology from gambling.